title	Mathematics Research Landscape
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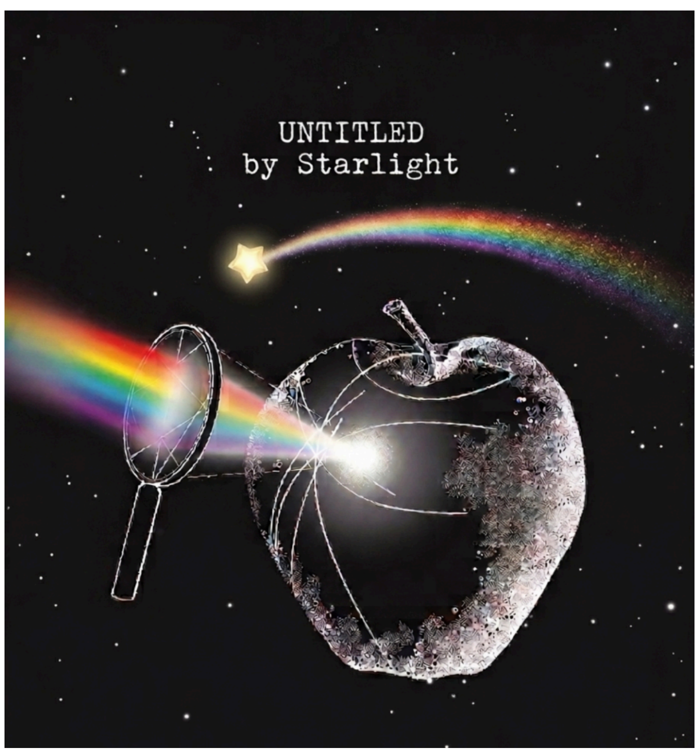
purpose	This document provides a research survey of LLM skill packs, Python algebraic solvers, local fine-tuned deep learning models, and empirical data construction strategies for solving the map from data to analytic expressions.
presumptive users	Engineers and researchers who need a landscape overview of tools and methods for symbolic regression, equation discovery, and computer algebra — without external API dependencies.
design principles	- undersell not oversell - machine readable for LLMs - comprehensive for replication

Note: These implementation guidelines and methodologies are subject to further development as our project standards evolve.



Insight: A Thesis On Theses And Generation

Life & Science In The Era of Faster Than The Speed Of You

From Data to Analytic Expressions: Research Landscape

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On A Post-IT

This document surveys the 2025–2026 landscape for going from raw data to closed-form analytic expressions — covering LLM-driven equation discovery systems (LLM-SR, SR-Scientist), local symbolic algebra libraries (SymPy, SageMath, Z3, PySR), fine-tuned deep learning models for mathematical reasoning (Qwen2.5-Math, DeepSeek-R1), and the core algorithms (genetic programming, SINDy, neural SR, dimensional analysis) with empirical data construction strategies. The emphasis is on fully local, no-external-API tooling that can run on consumer hardware.

Focuses

- **LLM orchestration**: Modern equation discovery uses LLMs as hypothesis proposers, backed by symbolic solvers for verification — not raw LLM math.
- **Local-first**: All listed tools run locally with no external API calls; SageMath and Z3 provide Wolfram-class symbolic capability.
- **Fine-tuning on consumer GPUs**: Qwen2.5-Math-1.5B and DeepSeek-R1 distilled fit on single GPUs via QLoRA/Unsloth.
- **Empirical rigor**: Benchmark construction (SRBench, LLM-SRBench) and data strategies (noise injection, domain splits, transform tests) prevent overclaimed results.

Design Focus

- **Section ordering**: LLM skill packs first because they represent the newest paradigm; traditional CAS libraries second as the foundation; fine-tuned models third as the emerging middle ground; core math last as reference. This lets a reader scan the bleeding edge first and drill down.
- **Flat bibliography**: All 55 references are in a single numbered list, grouped by type (papers, repos, PyPI, docs, search queries), so every source is citable by number.
- **No API-dependency assumption**: The brief explicitly calls out "no external APIs" because Wolfram/Mathematica/Maple are proprietary cloud-linked systems; the alternatives here are all local-first.

1. LLM Skill Packs for Equation Discovery & Symbolic Regression

The field has moved from "LLMs do math" to "LLMs orchestrate equation discovery." These systems all use an LLM to propose equation hypotheses, then verify and fit them with symbolic solvers.

LLM-SR (ICLR 2025 Oral) [bib]

- **Strategy:** LLM proposes equation skeletons as Python programs with placeholder parameters. Evolutionary search mutates and crosses over candidates. BFGS or Adam optimizers fit the constants against data. An experience buffer stores successful equations as in-context examples for the next iteration.
- **Backbone LLMs:** GPT-3.5-turbo, Mixtral-8x7B (works with open models locally).
- **Paper:** `arxiv.org/abs/2404.18400`
- **Code:** `github.com/deep-symbolic-mathematics/LLM-SR`

LLM-SRBench (ICML 2025 Oral) [bib]

- A comprehensive benchmark of 239 problems across four scientific domains designed specifically to prevent trivial LLM memorization.
- **Two categories:**
    - **LSR-Transform:** Common physical models rewritten into less common forms.
    - **LSR-Synth:** Synthetic discovery problems requiring genuine data-driven reasoning.
- **Code:** `github.com/deep-symbolic-mathematics/llm-srbench`
- **Dataset:** `huggingface.co/datasets/nnheui/llm-srbench`

SR-Scientist (ICLR 2026) [bib]

- Elevates the LLM from equation proposer to autonomous AI scientist. The agent writes code to analyze data, implements equations, submits them for evaluation, and iteratively improves based on experimental feedback.
- Uses a code interpreter toolset (data analysis, equation evaluation) over a long horizon with minimal human-defined pipelines.
- Includes an end-to-end reinforcement learning framework to enhance the agent.
- **Model:** SR-Scientist-30B on HuggingFace
- **Code:** `github.com/GAIR-NLP/SR-Scientist`

Other Notable Systems

System	Year	Approach
ProofOfThought [bib]	2025	LLM + Z3 theorem prover for verified reasoning chains
ICSR (In-Context Symbolic Regression) [bib]	2024	Iterative few-shot over candidate expressions with fitness feedback
LaSR [bib]	2024	Alternates hypothesis evolution, concept abstraction, and concept iteration

Common Architecture Pattern

```
[Raw Data] → [LLM proposes equation skeleton as code]
           → [SymPy simplifies / manipulates symbolically]
           → [BFGS/Adam fits numerical parameters]
           → [Evolutionary search explores variants]
           → [Dimensional analysis prunes invalid forms]
           → [Verifier checks fit + consistency]
           → [Output: closed-form analytic expression]
```

2. Python Libraries for Complex Algebraic Solving (Local, No External APIs)

Full Computer Algebra Systems

Library	Description	Install
SymPy [bib]	Mature symbolic CAS. Solve, simplify, integrate, differentiate, ODEs, linear algebra. Written entirely in Python.	▷ `pip install sympy`
SageMath [bib]	The most complete open-source Mathematica alternative. Bundles SymPy + Maxima + GAP + FLINT + NumPy/SciPy + R. Python-based language.	▷ `conda install sage`
Mathics3 [bib]	Mathematica-compatible syntax and functions. Built on SymPy. Includes Django web interface and terminal client.	▷ `pip install mathics`

Theorem Proving & SMT Solving

Library	Description	Install
Z3 [bib]	Microsoft's SMT solver. Supports arithmetic, bit-vectors, arrays, quantifiers, nonlinear constraints. Python bindings via `z3-solver`. 12.4k stars.	▷ `pip install z3-solver`
dReal [bib]	δ-complete SMT solver for nonlinear constraints over the reals. Handles transcendental functions.	Special build required

Symbolic Regression (Data → Expression)

Library	Approach	Install / Link
PySR [bib]	Genetic programming + simulated annealing + gradient-free constant optimization. Julia backend, Python API. The gold standard for practical SR.	▷ `pip install pysr`

Library	Approach	Install / Link
gplearn [bib]	Pure Python genetic programming for symbolic regression. Simpler, less performant than PySR.	▷ `pip install gplearn`
Operon [bib]	C++ genetic programming for SR with Python bindings. Very fast.	▷ `pip install operon`
AI Feynman [bib]	Dimensional analysis + brute-force search + neural network interpolation. Recovers equations from the Feynman Lectures.	▷ `pip install aifeynman`
EQL (Equation Learner) [bib]	Shallow neural network with arithmetic operator units (add, multiply, sin, etc.), trained end-to-end via backpropagation.	`github.com/martius-lab/EQL`
DeepSymRegTorch [bib]	PyTorch reimplementation of the MIT EQL network for integration with deep learning pipelines.	`github.com/samuelkim314/DeepSymRegTorch`

Dynamical System Identification

Library	Approach	Install
PySINDy [bib]	Sparse Identification of Nonlinear Dynamics. Builds a library of candidate functions, then uses LASSO/STLSQ to find the sparse governing equations.	▷ `pip install pysindy`
XL-SINDy [bib]	Lagrangian-based SINDy for robotics and mechanical systems. Discovers dynamics compatible with Lagrangian mechanics.	▷ `pip install py-xl-sindy`

Specialized Solvers

Library	Purpose	Install
mpmath [bib]	Arbitrary-precision arithmetic (included with SymPy)	▷ `pip install mpmath`
CVXPY [bib]	Convex optimization modeling	▷ `pip install cvxpy`
DEAP [bib]	Distributed evolutionary algorithms framework (used by many SR implementations)	▷ `pip install deap`

3. Local Fine-Tuned Deep Learning Models

Models You Can Run Locally

Model	Params	Domain
Qwen2.5-Math-1.5B [bib]	1.5B	Math reasoning, arithmetic, algebra, competition problems. Fits on consumer GPUs.
Qwen2.5-Math-7B [bib]	7B	Same, higher accuracy. Requires 24GB+ VRAM with quantization.
DeepSeekMath-V2 [bib]	7B/67B	Theorem proving with self-verification. Top on IMO/CMO/Putnam benchmarks.
DeepSeek-R1 (distilled) [bib]	1.5B–70B	Deep reasoning via chain-of-thought. Distilled versions run locally.
SR-Scientist-30B [bib]	30B	Fine-tuned specifically for scientific equation discovery in an agentic loop.
Llama 3.1/4 (fine-tuned) [bib]	8B–405B	General math after SFT on GSM8K/MATH/OpenMathReasoning datasets.

Fine-Tuning Infrastructure

Tool	Purpose
Unsloth [bib]	2x faster fine-tuning, 70% less VRAM. Supports Qwen3, Llama 4, DeepSeek. Native 128K context.
QLoRA (bitsandbytes) [bib]	4-bit quantized fine-tuning on consumer GPUs (e.g., Qwen2.5-Math-1.5B on 1xRTX 3090).
Axolotl [bib]	Configurable fine-tuning framework for open-source LLMs.
RLSF (RL via Symbolic Feedback) [bib]	Novel paradigm: uses Z3/SymPy as verifiers to provide token-level correctness signals during RL fine-tuning.

Standard Fine-Tuning Recipe for Math Reasoning

1. **Base model:** Qwen2.5-Math [bib] or DeepSeek-Math [bib] (smallest that fits your hardware)
2. **Dataset:** OpenMathReasoning [bib] (NVIDIA, millions of samples), GSM8K, MATH, or custom synthetic equation data
3. **Method:** SFT (Supervised Fine-Tuning) → optionally RL with verifier feedback
4. **Hardware:** 1.5B on 1xRTX 3090 with QLoRA [bib]; 7B on 1xA100; 30B+ on multi-GPU

4. Core Mathematics & Empirical Data Construction

Algorithmic Families for Expression Discovery

Genetic Programming (GP)

Evolves expression trees via crossover and mutation. Libraries: PySR [bib], gplearn [bib], Operon [bib], DEAP [bib].

- **Strengths:** Basis-free, can discover arbitrary functional forms.
- **Weaknesses:** Slow on high-dimensional data; constant optimization is hard.
- **Maturity:** PySR is production-ready with simulated annealing + multi-population evolution.

Sparse Regression (SINDy)

Builds a library Θ of candidate functions (polynomials, trig, etc.), then solves `Xdot = Θ(X)Ξ` with a sparsity penalty to find the few active terms. [bib]

- **Strengths:** Very fast, provably consistent under restricted-isometry conditions.
- **Weaknesses:** Limited to functions in the library; needs clean derivatives.

Neural Symbolic Regression

- **EQL (Equation Learner):** [bib] Replaces hidden layer activations with arithmetic operators (+, ×, sin, exp). Trained via standard backpropagation.
- **Transformers:** Pre-trained on millions of synthetic equations, can predict expressions from data in a single forward pass. Examples: NeSymReSo [bib], SymbolicGPT [bib], End-to-End Symbolic Regression [bib].
- **Strengths:** Fast inference after training; can learn strong priors.
- **Weaknesses:** Requires massive pretraining; generalization beyond training distribution is uncertain.

LLM-Guided (Hybrid)

LLM proposes equation skeletons, optimizer fits parameters, evolutionary search explores variations. Systems: LLM-SR [bib], SR-Scientist [bib].

- **Strengths:** Leverages scientific domain knowledge; flexible representations (equations as code, not just trees).
- **Weaknesses:** LLM cost per iteration; can hallucinate.

Dimensional Analysis (AI Feynman)

Use physical units (time, length, mass) to prune the search space. Only dimensionally consistent expressions survive, dramatically reducing the combinatorics. [bib]

Empirical Data Construction Strategies

From SR4Real, LLM-SRBench [bib], and SRBench [bib] methodologies:

Strategy	Description
Base	Sample equations from known physical laws (Feynman Lectures, etc.), generate (X, y) tuples with controlled noise
Transform	Modify equations via substitution/composition/rewriting — tests if model reasons or just recalls
Synthetic	Generate completely artificial symbolic expressions with random operators
Domain splits	In-distribution vs. out-of-distribution data to measure generalization
Noise injection	Add Gaussian noise at varying SNR ratios (1%, 5%, 10%)
Data scarcity	Evaluate at small-sample regimes (50–500 points)
SRBench	Standardized 252-problem benchmark across 14 scientific domains (Feynman, ODEs, etc.)

Benchmark Datasets

Benchmark	# Problems	Domain	Year
SRBench (Feynman) [bib]	119	Physics equations	2021
SRBench (Black-box) [bib]	133	Unknown ground-truth	2021
SRSD [bib]	80	Scientific discovery	2022
LLM-SRBench [bib]	239	Physics + biology + materials + synthetic	2025
AI Feynman I/II/III [bib]	100+	Physics with increasing complexity	2020/2025

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17. `github.com/SJ001/AI-Feynman` — AI Feynman symbolic regression
18. `github.com/martius-lab/EQL` — Equation Learner neural network
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20. `github.com/dynamicslab/pysindy` — Sparse identification of nonlinear dynamics
21. `github.com/z3prover/z3` — Z3 theorem prover (12.4k stars)
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Libraries on PyPI

23. `pypi.org/project/sympy` — SymPy symbolic mathematics
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Search Queries Used in This Research

1. Python libraries symbolic regression data to analytic expressions 2025 2026
2. local open source symbolic algebra solver Python like Wolfram Mathematica 2025
3. fine-tuned LLM symbolic mathematics equation discovery deep learning 2025 2026
4. LLM skill packs tools symbolic computation mathematical reasoning Python 2025
5. PySR symbolic regression Python library Miles Cranmer 2025
6. Z3 theorem prover Python SMT solver symbolic algebra 2025
7. AI Feynman symbolic regression neural network deep learning equation discovery 2025 2026
8. fine-tune local LLM mathematical reasoning symbolic computation Python library 2025
9. EQL equation learner neural network symbolic regression PyTorch implementation
10. SINDy sparse identification nonlinear dynamics Python library 2025

Changelog

- **v0.0.4** — added cover image (`untitled.simple.book.jacket.v0.0.4.png`) and new page break before title section.
- **v0.0.3** — converted all URLs to clickable hyperlinks; added HTML anchors and `[bib]` citation links throughout sections 1–4; expanded bibliography from 41 to 55 entries with coverage for all cited tools and models.
- **v0.0.2** — added Starlight-standard header table, On A Post-IT summary, Focuses, Design Focus, and Changelog per standards STARLIGHT for DOCS.
- **v0.0.1** — initial version: research landscape covering LLM skill packs, Python algebraic solvers, fine-tuned models, and empirical data construction; compiled via deep web research with 41 bibliographic entries.